| Format | Instruction | Opcode | Funct3 | Funct6/7 |
|---|---|---|---|---|
| R-type | add | 0110011 | 000 | 0000000 |
| | sub | 0110011 | 000 | 0100000 |
| | sll | 0110011 | 001 | 0000000 |
| | xor | 0110011 | 100 | 0000000 |
| | srl | 0110011 | 101 | 0000000 |
| | sra | 0110011 | 101 | 0000000 |
| | or | 0110011 | 110 | 0000000 |
| | and | 0110011 | 111 | 0000000 |
| | lr.d | 0110011 | 011 | 0001000 |
| | sc.d | 0110011 | 011 | 0001100 |
| I-type | lb | 0000011 | 000 | n.a. |
| | lh | 0000011 | 001 | n.a. |
| | lw | 0000011 | 010 | n.a. |
| | ld | 0000011 | 011 | n.a. |
| | lbu | 0000011 | 100 | n.a. |
| | lhu | 0000011 | 101 | n.a. |
| | lwu | 0000011 | 110 | n.a. |
| | addi | 0010011 | 000 | n.a. |
| | slli | 0010011 | 001 | 000000 |
| | xori | 0010011 | 100 | n.a. |
| | srli | 0010011 | 101 | 000000 |
| | srai | 0010011 | 101 | 010000 |
| | ori | 0010011 | 110 | n.a. |
| | andi | 0010011 | 111 | n.a. |
| | jalr | 1100111 | 000 | n.a. |
| S-type | sb | 0100011 | 000 | n.a. |
| | sh | 0100011 | 001 | n.a. |
| | sw | 0100011 | 010 | n.a. |
| | sd | 0100011 | 111 | n.a. |
| SB-type | beq | 1100111 | 000 | n.a. |
| | bne | 1100111 | 001 | n.a. |
| | blt | 1100111 | 100 | n.a. |
| | bge | 1100111 | 101 | n.a. |
| | bltu | 1100111 | 110 | n.a. |
| | bgeu | 1100111 | 111 | n.a. |
| U-type | lui | 0110111 | n.a. | n.a. |
| UJ-type | jal | 1101111 | n.a. | n.a. |

| Name | Field | | | | | | Comments |
|---|---|---|---|---|---|---|---|
| (Field Size) | 7 bits | 5 bits | 5 bits | 3 bits | 5 bits | 7 bits | |
| R-type | funct7 | rs2 | rs1 | funct3 | rd | opcode | Arithmetic instruction format |
| I-type | immediate[11:0] | | rs1 | funct3 | rd | opcode | Loads & immediate arithmetic |
| S-type | immed[11:5] | rs2 | rs1 | funct3 | immed[4:0] | opcode | Stores |
| SB-type | immed[12,10:5] | rs2 | rs1 | funct3 | immed[4:1,11] | opcode | Conditional branch format |
| UJ-type | immediate[20,10:1,11,19:12] | | | | rd | opcode | Unconditional jump format |
| U-type | immediate[31:12] | | | | rd | opcode | Upper immediate format |